

MERGE SORT:

Merge Sort is a divide-and-conquer sorting algorithm. It divides the array into smaller parts, sorts them, and then merges the sorted parts.

ALGORITHM:

1. Divide the array into two halves.
2. Continue dividing until each part contains one element.
3. Compare elements of the divided parts.
4. Merge them in sorted order.
5. Continue merging until the complete array is sorted.

Time Complexity:

Case	Complexity
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Best	$O(n \log n)$
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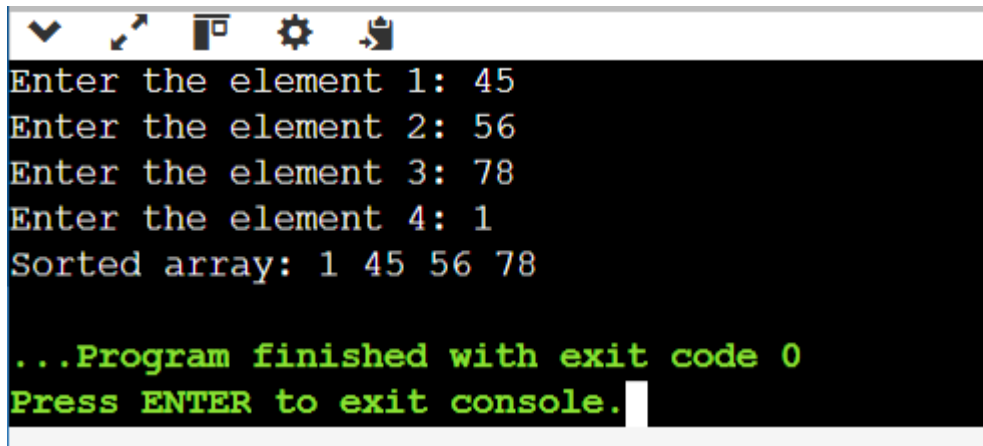
Average	$O(n \log n)$
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Worst	$O(n \log n)$
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Space Complexity:

Space Complexity = $O(n)$

OUTPUT:



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Enter the element 1: 45
Enter the element 2: 56
Enter the element 3: 78
Enter the element 4: 1
Sorted array: 1 45 56 78

...Program finished with exit code 0
Press ENTER to exit console.
```

CONCLUSION:

Merge Sort provides a guaranteed **$O(n \log n)$** time complexity in all cases. It is reliable for large datasets, but it requires extra memory for the merging process.